

Blueprint ka computational hissa execute hua

Seedhi baat: haan, software wala kaam chal raha hai; full fracture validation abhi nahi.

AB TAK KYA PASS HUA

Thermal code ke energy-balance, mesh/time refinement aur insulated-limit checks pass hue.

Thermal model ke outputs

Public APDL ke h, radius aur thermal properties par 100 ms ka reduced radial calculation chalaya. Shock input 100 K APDL default tha; yeh ANSYS reproduction ya fracture prediction nahi hai.

Case	Center C	Surface C	Model
Alumina monolith	99.685	45.068	radial FV
ZTA monolith	106.703	40.180	radial FV
ZTA core + 0.4 mm Al shell	105.458	41.794	radial FV*

*Shell thickness 0.4 mm article geometry se li gayi hai. Archived APDL geometry ise zero kar deti hai, isliye yeh independent thermal case hai, APDL run ka reproduction nahi.

Aur kaun se checks hue?

- Shao sphere script rerun: 219.27 K, 548.32 K, 1290.95 K ke existing thresholds reproduce hue; comparison ab bhi coarse experimental text limits se hai.
- Wang Appendix A algebra check rerun: published Eq. (A7) inconsistency wahi mili; yeh full PF-CZM solver test nahi hai.

Full fracture run kyun nahi kiya?

Source files me kuch settings abhi uniquely traceable nahi hain.

Reproduction se pehle 4 blockers

- Geometry: rod_height absent ho to APDL 5 mm default karta hai; article rod length 50 mm batata hai. Run command me 50 mm ka override documented nahi mila.
- Layer: geometry file layer_thickness = 0 um set karti hai; 400 um shell ke liye source edit/override chahiye.
- Temperature properties: 20 C row active hai; 320/620 C ke E aur expansion data commented out hain.
- ZTA strength: APDL = 902 MPa, Weibull m=13.6; article Table 1 = 1025 MPa, m=5.3. Conversion/definition abhi unresolved hai.
- Archive: metadata me CSV listed hai, par official preview HTTP 500 deta raha; ZIP/checksum locally verify nahi hue.

Input sanity checks (not FEM)

Material	Diffusivity m2/s	Biot number	Gc J/m2
Alumina	1.0202e-5	3.6765	25.403
ZTA	7.6140e-6	5.6818	46.181

Interpretation

Yeh differences paper galat hone ka proof nahi hain - kuch settings interactive ya undocumented ho sakti hain. Lekin guessed inputs se kiya run defensible reproduction nahi hoga, isliye fracture solver ko abhi rokna sahi hai.

Agla kaam aur clear boundary

Execution status ko honestly separate rakha hai: kya run hua, kya pending hai.

Computational next steps

- Official Zenodo archive/CSV ko readable download path se verify karna; phir material and parameter provenance lock karna.
- Documented geometry/settings milne ke baad independent axisymmetric thermoelastic and crack-energy model banana; mesh/time/crack-step convergence dikhana.
- Shao aur Papsik historical checks ko new prospective test se alag report karna. Published photos blind test nahi hain.

Blueprint ki mandatory validation

Composite architecture ko predictively validate karne ke liye blueprint me prospective, blinded rod experiment rakha hai. Iske liye actual specimens, measured material properties, instrumented quench, aur lab ki zaroorat hai; yeh code repository se conduct nahi ho sakta.

Abhi ke guardrails

No post-shock fitting. No author data request. No 3D simulation. Koi result Wang (2024) PF-CZM/FEM solver ko validate nahi karta. Q1 journal aim hai, acceptance guarantee nahi.

Files

- plan_b_blueprint.md - complete research protocol and go/no-go gates.
- plan_b_execution_log.md - executed checks, numerical outputs, and blockers.
- papsik_radial_thermal.py - reduced thermal model; papsik_input_audit.py - input scale checks.